PLC-BASED CONVEYOR AUTOMATION

Industrial Controls Engineering Portfolio | Siemens S7-1200 + WinCC HMI | TIA Portal V13 SP2

PROJECT OVERVIEW A simulation-based conveyor control application integrating PLC ladder logic, operator HMI, animated product movement, product counting, operating modes, fault detection and recovery. Developed to demonstrate practical controls design, commissioning, and root-cause troubleshooting.

01 / Technology Stack

Component	Platform / Role
Controller	Siemens S7-1200 CPU 1214C DC/DC/DC
Engineering	TIA Portal V13 SP2; ladder logic (LAD)
HMI	WinCC Basic V13; SIMATIC KTP700 Basic PN
Validation	S7-PLCSIM and HMI Runtime Simulator

02 / Control Strategy

Automatic operation: Start/Stop control with motor latch, fault permissives and production target stop. The product counter increments from simulated sensor detections and stops the conveyor at a configured target of 10.

Manual operation: Separate mode selection with RUN, STOP and JOG controls. Mode status is shown on the Manual Control screen.

Animated material flow: HMI horizontal movement uses **BOX_POSITION (%MW30)** driven by **MOVE_PULSE (%M11.0)**. Movement increment was changed from +1 to +2 to improve visual speed in simulation.

03 / HMI Design

Screen	Operator Functions
Root / Home	Conveyor animation, START/STOP/RESET, product count, motor and sensor indicators, system status
Manual Control	AUTO/MANUAL selection, RUN/STOP/JOG, system interlocks and fault diagnostics
Alarms	Production target, sensor timeout and motor fault event visibility

Consistent HOME / MANUAL / ALARMS navigation was implemented across screens. HMI indicators were linked to PLC state tags for real-time simulation feedback.

04 / Fault Detection and Recovery

Motor fault: MOTOR_FAULT_SIM (%M16.4) is used to test a latched MOTOR_FAULT (%M16.5). The motor stops, an alarm bit is raised, and fault reset clears the latch after the simulation input is removed. The motor remains off until a new START command.

Sensor fault: SENSOR_FAIL_SIM (%M16.3) suppresses simulated detection. A TON-based sensor watchdog (12 s preset) can trigger SENSOR_TIMEOUT (%M16.0) and CONVEYOR_FAULT (%M16.2), stopping the conveyor.

Design limitation: The sensor timeout was observed to occur after the animated box had already passed the expected sensor area. A position-window-based missed-detection strategy has been proposed but has **not yet been implemented or validated**.

05 / Engineering Troubleshooting Highlight

During HMI navigation testing, switching from Manual Control to the Root screen unexpectedly activated both AUTO_MODE and MANUAL_MODE, preventing motor startup. Investigation isolated an unintended **SetTag(MANUAL_MODE, 1)** function in the Root screen **Loaded** event. Removing the function restored mutually exclusive mode selection and normal conveyor startup.

06 / Simulation Validation

Test Scenario	Observed / Confirmed Outcome
AUTO mode and navigation	Single selected mode remains active after screen transitions
Conveyor START/STOP	Motor starts and stops through HMI commands
Manual RUN/STOP/JOG	Reported functional during project testing
Production target	Counter reaches 10; conveyor stops and target alarm activates
Motor fault recovery	Fault simulation stops motor; reset clears fault without automatic restart
Sensor failure	Sensor fault observed; response time requires optimization
HMI build	Previously compiled with zero errors; noncritical warnings remained

07 / Key Engineering Skills Demonstrated

PLC ladder programming • Boolean interlocks • TON timers and counters • HMI screen development • Dynamic animations • Alarm configuration • Mode management • PLCSIM testing • Fault isolation • Functional validation • Root-cause analysis

08 / Next Engineering Iterations

Optimize sensor fault detection against the expected product position window; verify the motor fault alarm screen behavior in all operating scenarios; perform a final integrated HMI regression test; record conveyor operation, production target stop, and fault recovery for a portfolio demo.

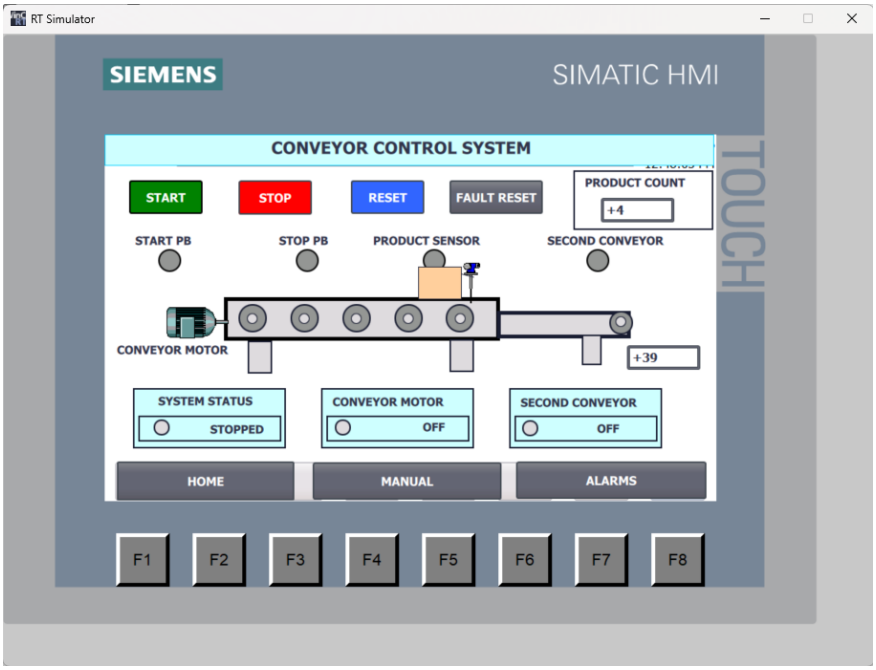
Project Scope & Evidence

This report documents a **simulated controls application**, not a deployed production system. Project descriptions and test outcomes are based on the development and testing history supplied in this conversation. The uploaded **conveyor_test.info** contains only metadata and does not independently verify the PLC program.

VISUAL EVIDENCE | ENGINEERING VALIDATION

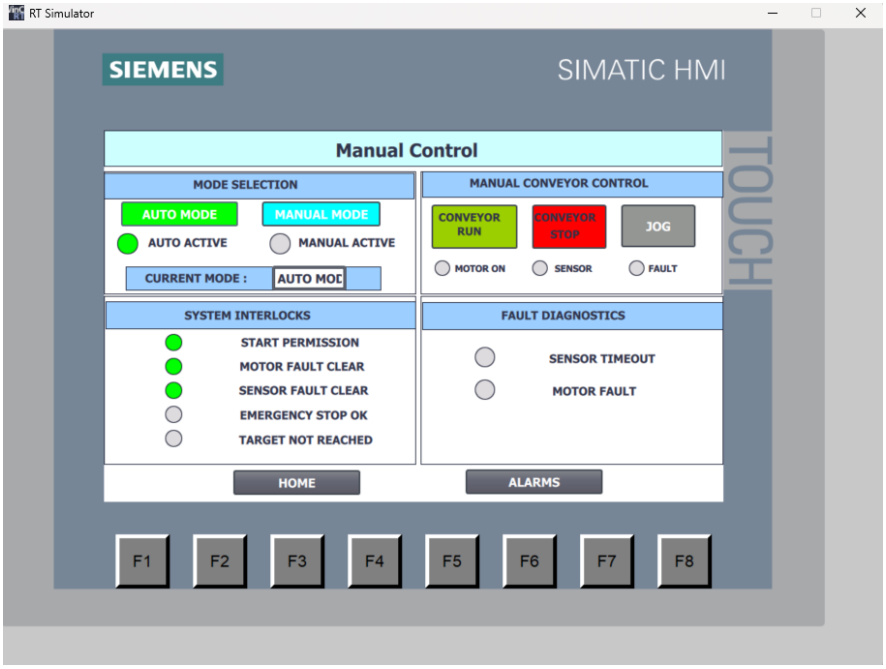
Original Siemens TIA Portal, WinCC Runtime and PLCSIM screenshots. All evidence is simulation-based; no physical commissioning is claimed.

01 | HMI - Conveyor overview



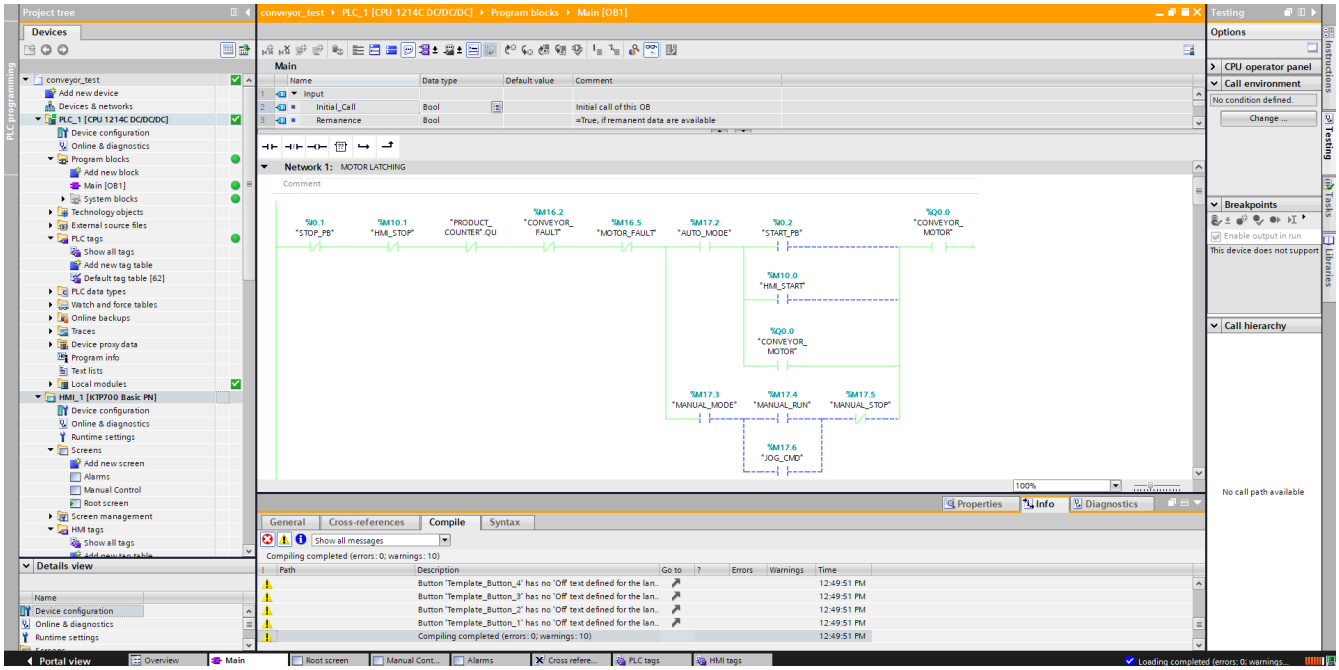
Operator interface shows conveyor visualization, product count, motor status, and navigation.

02 | HMI - Manual control and diagnostics



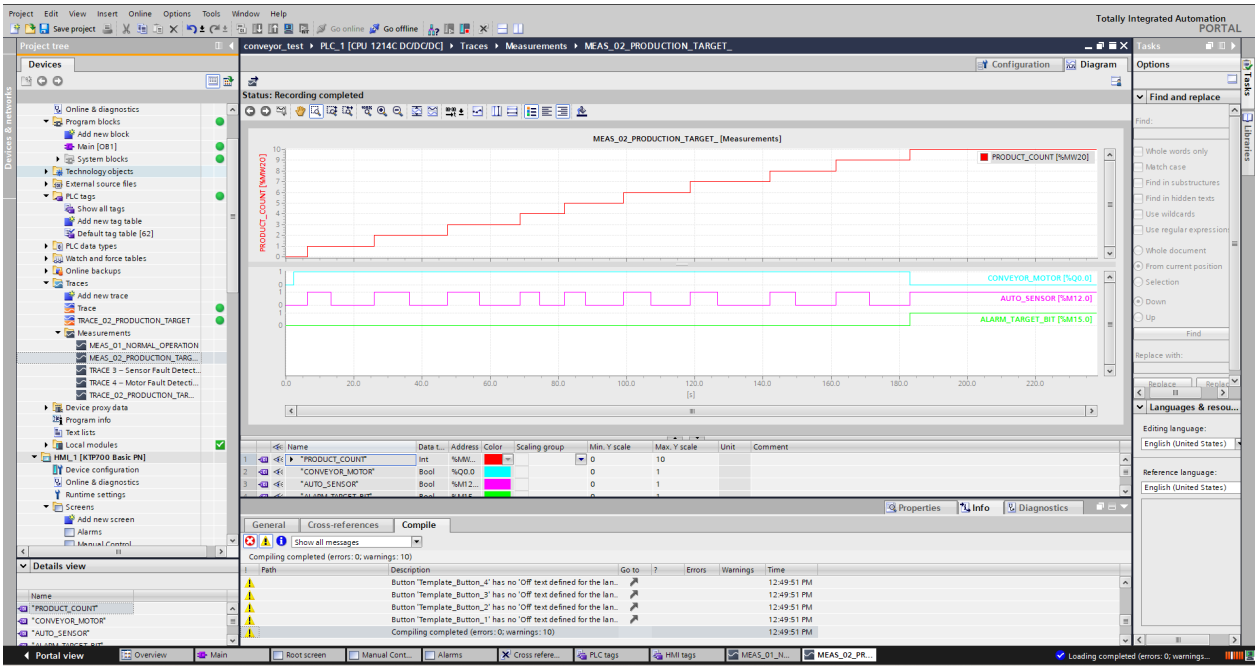
Automatic/manual selection, RUN/STOP/JOG controls, interlocks, and fault indicators.

03 | PLC - Motor control and interlocks



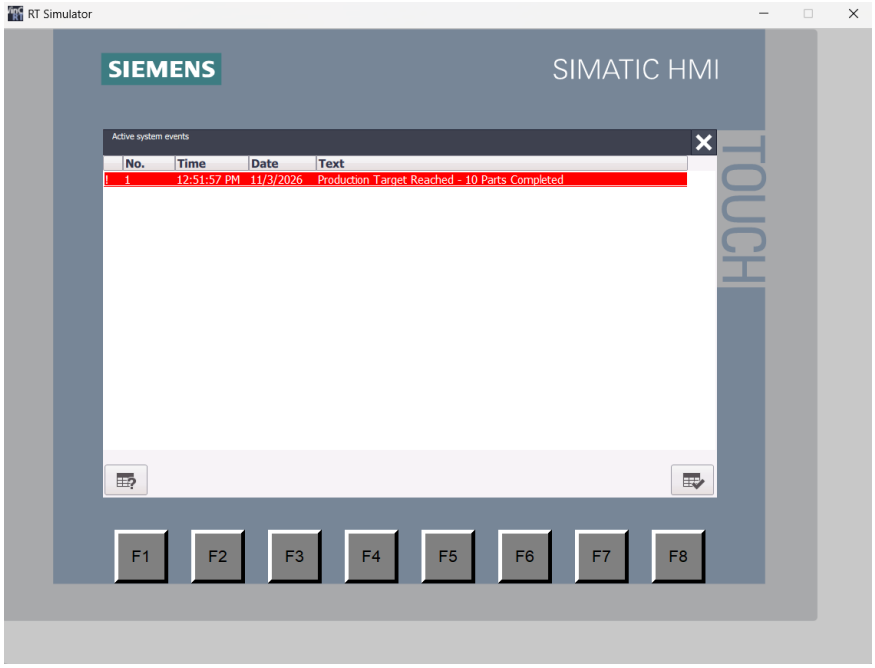
Ladder network shows motor latching, mode selection branches, and fault/target interlocks.

04 | Trace - Production target reached



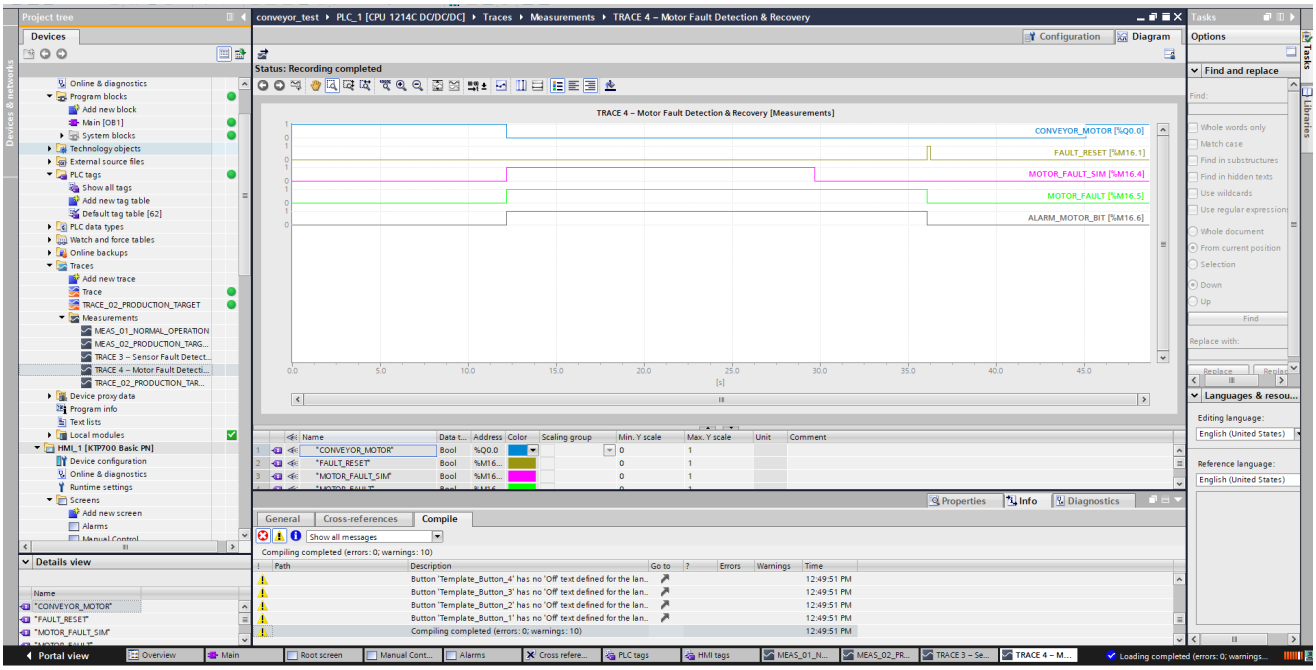
Product count increases to 10, target alarm activates, and conveyor motor stops.

05 | HMI - Production target alarm



Runtime alarm confirms: Production Target Reached - 10 Parts Completed.

06 | Trace - Motor fault and recovery



Motor fault simulation stops the conveyor; latched fault and alarm clear on reset.